

Chapter 3: Understanding users



Overview

- What is cognition?
- What are users good and bad at?
- Describe how cognition has been applied to interaction design
- Theories of cognition
 - Mental models, theory of action
 - Information processing
 - External cognition, distributed cognition

Why do we need to understand users?

- Interacting with technology is cognitive
- We need to take into account cognitive processes involved and cognitive limitations of users
- We can provide knowledge about what users can and cannot be expected to do
- Identify and explain the nature and causes of problems users encounter
- Supply theories, modelling tools, guidance and methods that can lead to the design of better interactive products

What goes on in the mind?

perceiving
thinking
remembering
learning

understanding others
talking with others
manipulating others

planning a meal
imagining a trip
painting
writing
composing



making decisions
solving problems
daydreaming

Core cognitive aspects

- Attention
- Perception and recognition
- Memory
- Reading, speaking and listening
- Problem-solving, planning, reasoning and decision-making, learning
- Most relevant to interaction design are attention, perception and recognition, and memory

Attention

- Selecting things to concentrate on at a point in time from the mass of stimuli around us
- Allows us to focus on information that is relevant to what we are doing
- Involves audio and/or visual senses
- Focussed and divided attention enables us to be selective in terms of the mass of competing stimuli but limits our ability to keep track of all events
- Information at the interface should be structured to capture users' attention, e.g. use perceptual boundaries (windows), colour, reverse video, sound and flashing lights

Activity: Find the price of a double room at the Holiday Inn in Bradley

Pennsylvania
Bedford Motel/Hotel: Crinaline Courts
(814) 623-9511 S: \$18 D: \$20
Bedford Motel/Hotel: Holiday Inn
(814) 623-9006 S: \$29 D: \$36
Bedford Motel/Hotel: Midway
(814) 623-8107 S: \$21 D: \$26
Bedford Motel/Hotel: Penn Manor
(814) 623-8177 S: \$19 D: \$25
Bedford Motel/Hotel: Quality Inn
(814) 623-5189 S: \$23 D: \$28
Bedford Motel/Hotel: Terrace
(814) 623-5111 S: \$22 D: \$24
Bradley Motel/Hotel: De Soto
(814) 362-3567 S: \$20 D: \$24
Bradley Motel/Hotel: Holiday House
(814) 362-4511 S: \$22 D: \$25
Bradley Motel/Hotel: Holiday Inn
(814) 362-4501 S: \$32 D: \$40
Breezewood Motel/Hotel: Best Western Plaza
(814) 735-4352 S: \$20 D: \$27
Breezewood Motel/Hotel: Motel 70
(814) 735-4385 S: \$16 D: \$18

Activity: Find the price for a double room at the Quality Inn in Columbia

South Carolina

City	Motel/Hotel	Area code	Phone	Rates	
				Single	Double
Charleston	Best Western	803	747-0961	\$26	\$30
Charleston	Days Inn	803	881-1000	\$18	\$24
Charleston	Holiday Inn N	803	744-1621	\$36	\$46
Charleston	Holiday Inn SW	803	556-7100	\$33	\$47
Charleston	Howard Johnsons	803	524-4148	\$31	\$36
Charleston	Ramada Inn	803	774-8281	\$33	\$40
Charleston	Sheraton Inn	803	744-2401	\$34	\$42
Columbia	Best Western	803	796-9400	\$29	\$34
Columbia	Carolina Inn	803	799-8200	\$42	\$48
Columbia	Days Inn	803	736-0000	\$23	\$27
Columbia	Holiday Inn NW	803	794-9440	\$32	\$39
Columbia	Howard Johnsons	803	772-7200	\$25	\$27
Columbia	Quality Inn	803	772-0270	\$34	\$41
Columbia	Ramada Inn	803	796-2700	\$36	\$44
Columbia	Vagabond Inn	803	796-6240	\$27	\$30

Activity

- Tullis (1987) found that the two screens produced quite different results
 - 1st screen - took an average of 5.5 seconds to search
 - 2nd screen - took 3.2 seconds to search
- Why, since both displays have the same density of information (31%)?
- Spacing
 - In the 1st screen the information is bunched up together, making it hard to search
 - In the 2nd screen the characters are grouped into vertical categories of information making it easier

Design implications for attention

- Make information salient when it needs attending to
- Use techniques that make things stand out like colour, ordering, spacing, underlining, sequencing and animation
- Avoid cluttering the interface - follow the google.com example of crisp, simple design
- Avoid using too much because the software allows it

An example of over-use of graphics



Our Situation

- ◆ State the bad news
- ◆ Be clear, don't try to obscure the situation

Perception and recognition

- How information is acquired from the world and transformed into experiences
- Obvious implication is to design representations that are readily perceivable, e.g.
 - Text should be legible
 - Icons should be easy to distinguish and read

Is color contrast good? Find italian

Black Hills Forest Cheyenne River Social Science South San Jose Badlands Park Juvenile Justice	Peters Landing Public Health San Bernardino Moreno Valley Altamonte Springs Peach Tree City	Jefferson Farms Psychophysics Political Science Game Schedule South Addition Cherry Hills Village	Devlin Hall Positions Hubard Hall Fernadino Beach Council Bluffs Classical Lit
Results and Stats Thousand Oaks Promotions North Palermo Credit Union Wilner Hall	Highland Park Manchesney Park Vallecito Mts. Rock Falls Freeport Slaughter Beach	Creative Writing Lake Havasu City Engineering Bldg Sports Studies Lakewood Village Rock Island	Sociology Greek Wallace Hall Concert Tickets Public Radio FM Children's Museum
Performing Arts Italian Coaches McKees Rocks Glenwood Springs Urban Affairs	Rocky Mountains Latin Pleasant Hills Observatory Public Affairs Heskett Center	Deerfield Beach Arlington Hill Preview Game Richland Hills Experts Guide Neff Hall	Writing Center Theater Auditions Delaware City Scholarships Hendricksville Knights Landing
McLeansboro Experimental Links Graduation Emory Lindquist Clinton Hall San Luis Obispo	Brunswick East Millinocket Women's Studies Vacant News Theatre Candlewood Isle	Grand Wash Cliffs Indian Well Valley Online Courses Lindquist Hall Fisk Hall Los Padres Forest	Modern Literature Studio Arts Hughes Complex Cumberland Flats Central Village Hoffman Estates

Are borders and white space better? Find french

Webmaster
Russian
Athletics
Go Shockers
Degree Options
Newsletter

Curriculum
Emergency (EMS)
Statistics
Award Documents
Language Center
Future Shockers

Student Life
Accountancy
McKnight Center
Council of Women
Commute
Small Business

Dance
Gerontology
Marketing
College Bylaws
Why Wichita?
Tickets

Geology
Manufacturing
Management
UCATS
Alumni News
Saso

Intercollegiate
Bowling
Wichita Gateway
Transfer Day
Job Openings
Live Radio

Thinker & Movers
Alumni
Foundations
Corbin Center
Jardine Hall
Hugo Wall School

Career Services
Doers & Shockers
Core Values
Grace Wilkie Hall
Strategic Plan
Medical Tech

Educational Map
Physical Plant
Graphic Design
Non Credit Class
Media Relations
Advertising

Beta Alpha Psi
Liberal Arts
Counseling
Biological Science
Duerksen Fine Art
EMT Program

Staff
Aerospace
Choral Dept.
Alberg Hall
French
Spanish

Softball, Men's
McKinley Hall
Email
Dental Hygiene
Tenure
Personnel Policies

English
Graduate Complex
Music Education
Advising Center
Medical School
Levitt Arena

Religion
Art Composition
Physics
Entrepreneurship
Koch Arena
Roster

Parents
Wrestling
Philosophy
Wichita Lyceum
Fairmount Center
Women's Museum

Instrumental
Nursing
Opera
Sports History
Athletic Dept.
Health Plan

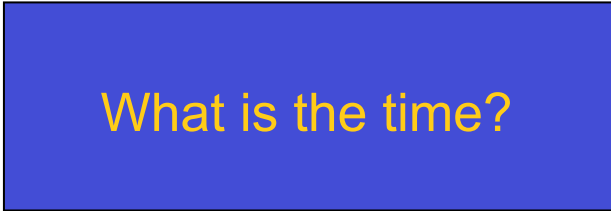
Activity

- Weller (2004) found people took less time to locate items for information that was grouped
 - using a border (2nd screen) compared with using color contrast (1st screen)
- Some argue that too much white space on web pages is detrimental to search
 - Makes it hard to find information
- Do you agree?

Which is easiest to read and why?



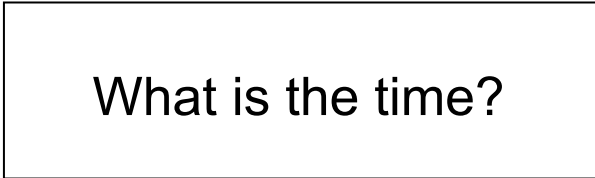
What is the time?



What is the time?



What is the time?



What is the time?



What is the time?

Design implications

- Representations of information need to be designed to be perceptible and recognizable
- Icons and other graphical representations should enable users to readily *distinguish* their meaning
- Bordering and spacing are effective visual ways of grouping information
- Sounds should be audible and distinguishable
- Speech output should enable users to distinguish between the set of spoken words
- Text should be legible and distinguishable from the background

Memory

- Involves first encoding and then retrieving knowledge
- We don't remember everything - involves filtering and processing what is attended to
- Context is important in affecting our memory (i.e., where, when)
- Well known fact that we recognize things much better than being able to recall things
 - Better at remembering images than words
 - Why interfaces are largely visual

Processing in memory

- Encoding is first stage of memory
 - determines which information is attended to in the environment and how it is interpreted
- The more attention paid to something,
- And the more it is processed in terms of thinking about it and comparing it with other knowledge,
- The more likely it is to be remembered
 - e.g., when learning about HCI, it is much better to reflect upon it, carry out exercises, have discussions with others about it, and write notes than just passively read a book, listen to a lecture or watch a video about it

Context is important

- Context affects the extent to which information can be subsequently retrieved
- Sometimes it can be difficult for people to recall information that was encoded in a different context
 - e.g., You are on a train and someone comes up to you and says hello. You don't recognize him for a few moments but then realize it is one of your neighbors. You are only used to seeing your neighbor in the hallway of your apartment block and seeing him out of context makes him difficult to recognize initially

Activity

- Try to remember the dates of your grandparents' birthday
- Try to remember the cover of the last two DVDs you bought or rented
- Which was easiest? Why?
- People are very good at remembering visual cues about things
 - e.g., the color of items, the location of objects and marks on an object
- They find it more difficult to learn and remember arbitrary material
 - e.g., birthdays and phone numbers

Recognition versus recall

- Command-based interfaces require users to recall from memory a name from a possible set of 100s
- GUIs provide visually-based options that users need only browse through until they recognize one
- Web browsers, MP3 players, etc., provide lists of visited URLs, song titles etc., that support recognition memory

The problem with the classic ' 7 ± 2 '

- George Miller's theory of how much information people can remember
- People's immediate memory capacity is very limited
- Many designers have been led to believe that this is useful finding for interaction design

What some designers get up to...

- Present only 7 options on a menu
- Display only 7 icons on a tool bar
- Have no more than 7 bullets in a list
- Place only 7 items on a pull down menu
- Place only 7 tabs on the top of a website page

– But this is wrong? Why?



Why?

- Inappropriate application of the theory
- People can scan lists of bullets, tabs, menu items till they see the one they want
- They don't have to recall them from memory having only briefly heard or seen them
- Sometimes a small number of items is good design
- But it depends on task and available screen estate

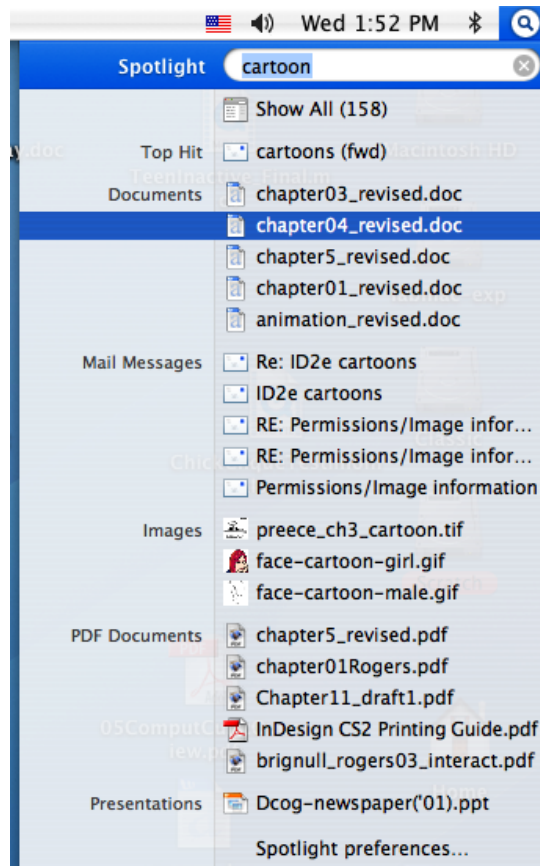
Personal information management

- Personal information management (PIM) is a growing problem for most users
 - Who have vast numbers of documents, images, music files, video clips, emails, attachments, bookmarks, etc.,
 - Major problem is deciding where and how to save them all, then remembering what they were called and where to find them again
 - Naming most common means of encoding them
 - Trying to remember a name of a file created some time back can be very difficult, especially when have 1000s and 1000s
 - How might such a process be facilitated taking into account people's memory abilities?

Personal information management

- Memory involves 2 processes
 - recall-directed and recognition-based scanning
- File management systems should be designed to optimize both kinds of memory processes
 - e.g., Search box and history list
- Help users encode files in richer ways
 - Provide them with ways of saving files using colour, flagging, image, flexible text, time stamping, etc

Is Apple's Spotlight search tool any good?



Design implications

- Don't overload users' memories with complicated procedures for carrying out tasks
- Design interfaces that promote recognition rather than recall
- Provide users with a variety of ways of encoding digital information to help them remember where they have stored them
 - e.g., categories, color, flagging, time stamping

Mental models

- Users develop an understanding of a system through learning and using it
- Knowledge is often described as a mental model
 - How to use the system (what to do next)
 - What to do with unfamiliar systems or unexpected situations (how the system works)
- People make inferences using mental models of how to carry out tasks

Mental models

- Craik (1943) described mental models as internal constructions of some aspect of the external world enabling predictions to be made
- Involves unconscious and conscious processes, where images and analogies are activated
- Deep versus shallow models (e.g. how to drive a car and how it works)

Everyday reasoning and mental models

- (a) You arrive home on a cold winter's night to a cold house. How do you get the house to warm up as quickly as possible? Set the thermostat to be at its highest or to the desired temperature?
- (b) You arrive home starving hungry. You look in the fridge and find all that is left is an uncooked pizza. You have an electric oven. Do you warm it up to 375 degrees first and then put it in (as specified by the instructions) or turn the oven up higher to try to warm it up quicker?

Heating up a room or oven that is thermostat-controlled

- Many people have erroneous mental models (Kempton, 1996)
- Why?
 - General valve theory, where 'more is more' principle is generalised to different settings (e.g. gas pedal, gas cooker, tap, radio volume)
 - Thermostats based on model of on-off switch model

Heating up a room or oven that is thermostat-controlled

- Same is often true for understanding how interactive devices and computers work:
 - Poor, often incomplete, easily confusable, based on inappropriate analogies and superstition (Norman, 1983)
 - e.g. elevators and pedestrian crossings - lot of people hit the button at least twice
 - Why? Think it will make the lights change faster or ensure the elevator arrives!

Exercise: ATMs

- Write down how an ATM works
 - How much money are you allowed to take out?
 - What denominations?
 - If you went to another machine and tried the same what would happen?
 - What information is on the strip on your card? How is this used?
 - What happens if you enter the wrong number?
 - Why are there pauses between the steps of a transaction? What happens if you try to type during them?
 - Why does the card stay inside the machine?
 - Do you count the money? Why?

How did you fare?

- Your mental model
 - How accurate?
 - How similar?
 - How shallow?
- Payne (1991) did a similar study and found that people frequently resort to analogies to explain how they work
- People's accounts greatly varied and were often ad hoc

Norman's (1986) Theory of action

- Proposes 7 stages of an activity
 - Establish a goal
 - Form an intention
 - Specify an action sequence
 - Execute an action
 - Perceive the system state
 - Interpret the state
 - Evaluate the system state with respect to the goals and intentions

An example: reading breaking news on the web

- (i) Set goal to find out about breaking news
decide on news website
- (ii) Form an intention
check out BBC website
- (iii) Specify what to do
move cursor to link on browser
- (iv) Execute action sequence
click on mouse button
- (v) Check what happens at the interface
see a new page pop up on the screen
- (vi) Interpret it
read that it is the BBC website
- (vii) Evaluate it with respect to the goal
read breaking news

How realistic?

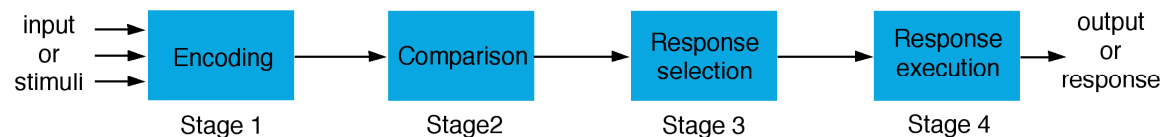
- Human activity does not proceed in such an orderly and sequential manner
- More usual for stages to be missed, repeated or out of order
- Do not always have a clear goal in mind but react to the world
- Theory is only approximation of what happens and is greatly simplified
- Help designers think about how to help users monitor their actions

The gulfs

- The 'gulfs' explicate the gaps that exist between the user and the interface
- The gulf of execution
 - the distance from the user to the physical system while the second one
- The gulf of evaluation
 - the distance from the physical system to the user
- Need to bridge the gulfs in order to reduce the cognitive effort required to perform a task

Information processing

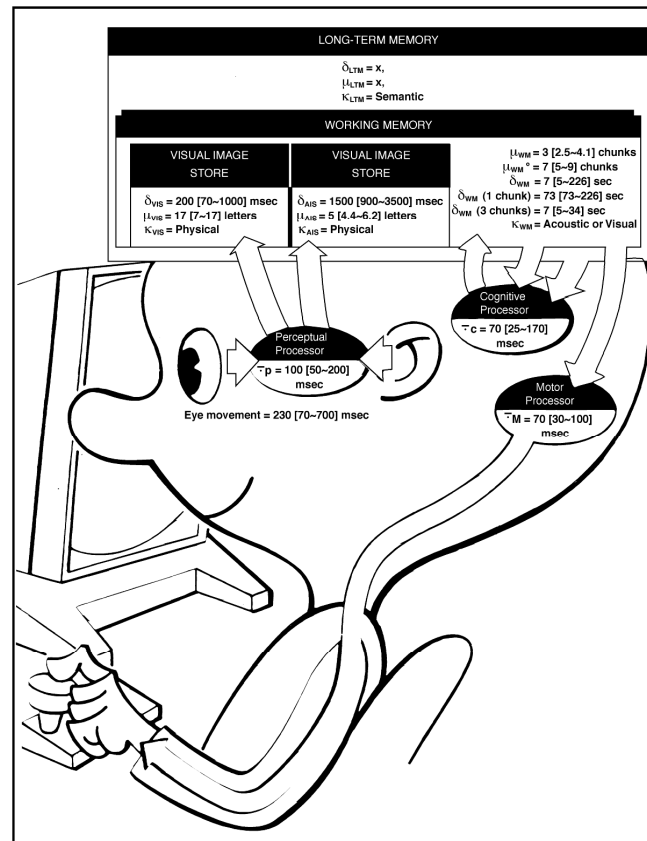
- Conceptualizes human performance in metaphorical terms of information processing stages



Model Human processor (Card et al, 1983)

- Models the information processes of a user interacting with a computer
- Predicts which cognitive processes are involved when a user interacts with a computer
- Enables calculations to be made of how long a user will take to carry out a task

The human processor model



External cognition

- Concerned with explaining how we interact with external representations (e.g. maps, notes, diagrams)
- What are the cognitive benefits and what processes involved
- How they extend our cognition
- What computer-based representations can we develop to help even more?

Externalizing to reduce memory load

- Diaries, reminders, calendars, notes, shopping lists, to-do lists - written to remind us of what to do
- Post-its, piles, marked emails - where placed indicates priority of what to do
- External representations:
 - Remind us that we need to do something (e.g. to buy something for mother's day)
 - Remind us of what to do (e.g. buy a card)
 - Remind us when to do something (e.g. send a card by a certain date)

Computational offloading

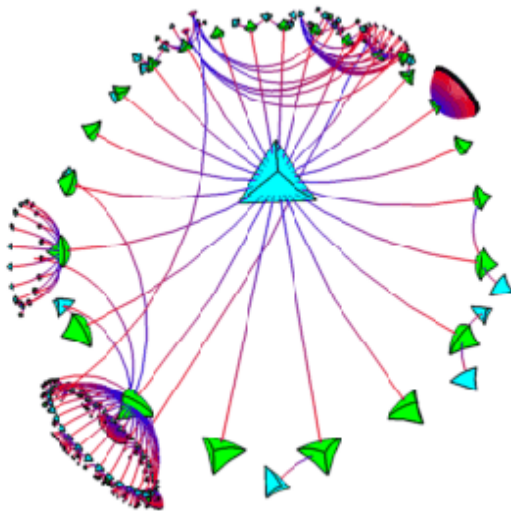
- When a tool is used in conjunction with an external representation to carry out a computation (e.g. pen and paper)
- Try doing the two sums below (a) in your head, (b) on a piece of paper and c) with a calculator.
 - $234 \times 456 = ??$
 - CCXXXIIII $\times$ CCCCXXXXXVI = ???
- Which is easiest and why? Both are identical sums

Annotation and cognitive tracing

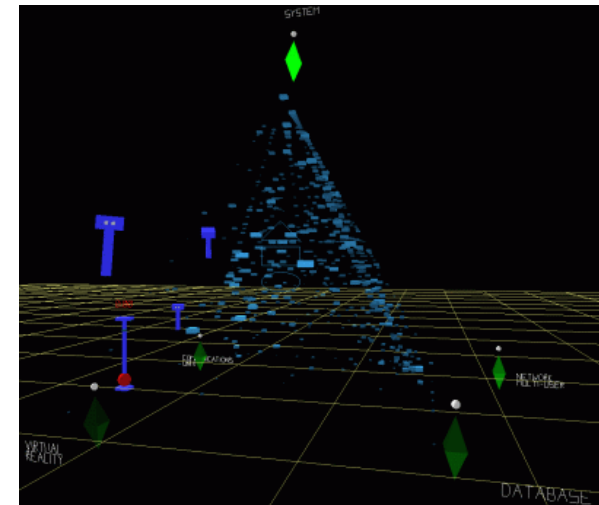
- Annotation involves modifying existing representations through making marks
 - e.g. crossing off, ticking, underlining
- Cognitive tracing involves externally manipulating items into different orders or structures
 - e.g. playing scrabble, playing cards

Design implication

- Provide external representations at the interface that reduce memory load and facilitate computational offloading



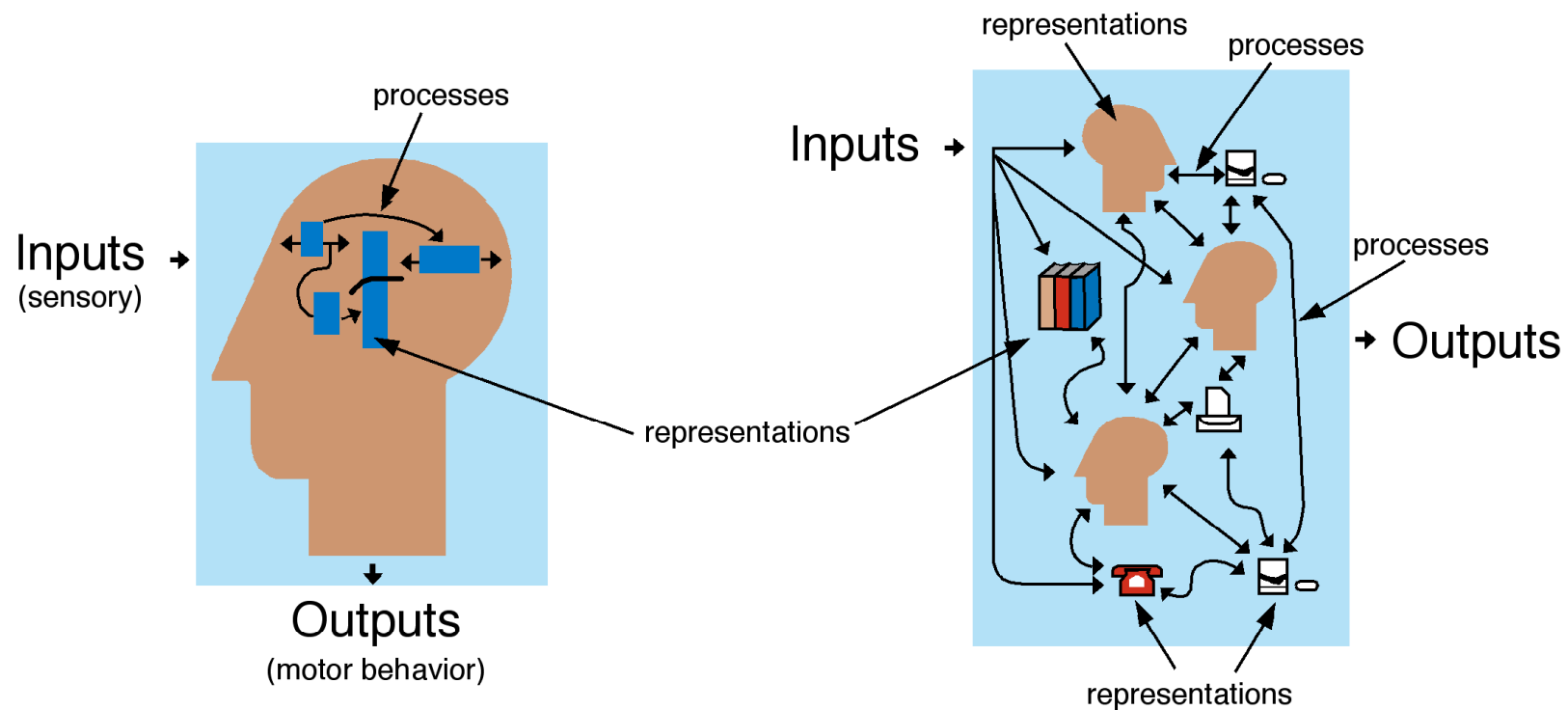
e.g. Information visualizations have been designed to allow people to make sense and rapid decisions about masses of data



Distributed cognition

- Concerned with the nature of cognitive phenomena across individuals, artifacts, and internal and external representations (Hutchins, 1995)
- Describes these in terms of propagation across representational state
- Information is transformed through different media (computers, displays, paper, heads)

How it differs from information processing



1. Traditional model

2. Distributed model

What's involved

- The distributed problem-solving that takes place
- The role of verbal and non-verbal behavior
- The various coordinating mechanisms that are used (e.g., rules, procedures)
- The communication that takes place as the collaborative activity progresses
- How knowledge is shared and accessed

Key points

- Cognition involves several processes including attention, memory, perception and learning
- The way an interface is designed can greatly affect how well users can perceive, attend, learn and remember how to do their tasks
- Theoretical frameworks such as mental models and external cognition provide ways of understanding how and why people interact with products, which can lead to thinking about how to design better products